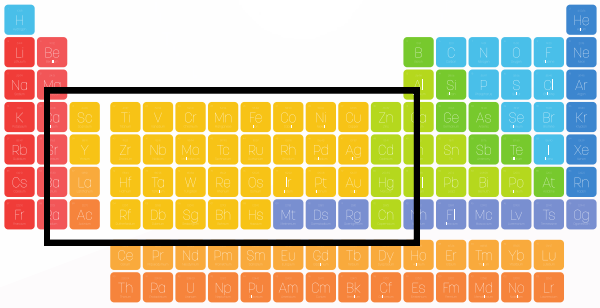
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success

JEE (MAIN+ADVANCED)

Inorganic Chemistry

ENTHUSIAST COURSE



STUDY MATERIAL

d- Block Compounds

English Medium

Copyright Statement

All rights including trademark and copyrights and rights of translation etc. reserved and vested exclusively with ALLEN Career Institute Private Limited. (ALLEN)

No part of this work may be copied, reproduced, adapted, abridged or translated, transcribed, transmitted, stored or distributed in any form retrieval system, computer system, photographic or other system or transmitted in any form or by any means whether electronic, magnetic, chemical or manual, mechanical, digital, optical, photocopying, recording or otherwise, or stood in any retrieval system of any nature without the written permission of the Allen Career Institute Private Limited. Any breach will entail legal action and prosecution without further notice.

This work is sold/distributed by Allen Career Institute Private Limited subject to the condition and undertaking given by the student that all proprietary rights (under the Trademark Act, 1999 and Copyright Act, 1957) of the work shall be exclusively belong to ALLEN Career Institute Private Limited. Neither the Study Materials and/or Test Series and/or the contents nor any part thereof i.e. work shall be reproduced, modify, re-publish, sub-license, upload on website, broadcast, post, transmit, disseminate, distribute, sell in market, stored in a retrieval system or transmitted in any form or by any means for reproducing or making multiple copies of it.

Any person who does any unauthorised act in relation to this work may be liable to criminal prosecution and civil claims for damages. Any violation or infringement of the propriety rights of Allen shall be punishable under Section- 29 & 52 of the Trademark Act, 1999 and under Section- 51, 58 & 63 of the Copyright Act, 1957 and any other Act applicable in India. All disputes are subjected to the exclusive jurisdiction of courts, tribunals and forums at Kota, Rajasthan only.

Note:- This publication is meant for educational and learning purposes. All reasonable care and diligence have been taken while editing and printing this publication. ALLEN Career Institute Private Limited shall not hold any responsibility for any error that may have inadvertently crept in.
ALLEN Career Institute Private Limited is not responsible for the consequences of any action taken on the basis of this publication.

d-BLOCK COMPOUND

TRANSITION ELEMENTS

Definition : They one often called 'transition elements' because their position in the periodic table is between s-block and p-block elements

Typically, the transition elements have incompletely filled d-level. Since Zn group has d^{10} configuration in their ground state as well as in stable oxidation state, they are not considered as transition elements but they ared-block elements.

1st Series

	Sc	Ti	V	**Cr**	Mn	Fe	Co	Ni	**Cu**	Zn
Z	21	22	23	24	25	26	27	28	29	30
4s	2	2	2	1	2	2	2	2	1	2
3d	1	2	3	5	5	6	7	8	10	10

2nd Series

	Y	Zr	**Nb**	**Mo**	**Tc**	**Ru**	**Rh**	**Pd**	**Ag**	Cd
Z	39	40	41	42	43	44	45	46	47	48
5s	2	2	1	1	2	1	1	0	1	2
4d	1	2	4	5	5	7	8	10	10	10

3rd Series

	La	Hf	Ta	W	Re	Os	Ir	**Pt**	**Au**	Hg
Z	57	72	73	74	75	76	77	78	79	80
6s	2	2	2	2	2	2	2	1	1	2
5d	1	2	3	4	5	6	7	9	10	10

4th Series

	Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds	**Rg**	Uub
Z	89	104	105	106	107	108	109	110	111	112
7s	2	2	2	2	2	2	2	2	1	2
6d	1	2	3	4	5	6	7	8	10	10

General Characteristics :

(i) **Metallic character :** They are all metal and good conductor of heat & electricity

(ii) **Electronic configuration :** $(n-1)d^{1-10}ns^{1-2}$

	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
others are as usual				$4s^1$ $3d^5$					$4s^1$ $3d^{10}$	

(iii) **M.P.** Cr, Mo, W ⟶ Maximum 6 no. of unpaired e^-s are involved in metallic bonding

Zn, Cd, Hg lowest m.p. due to no unpaired e^- for metallic bonding

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\6-Salt Anal+Heating Eff+s,d Block+Run of Salt Anal-(OC)\Eng\04_d-Block\01_Th+Ex.p65

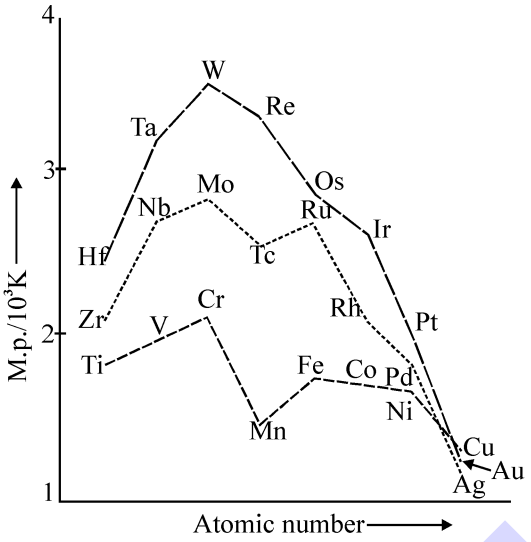


Fig. : Trends in melting points of transition elements

(iv) Variation in atomic radius :

Sc ⟶ Mn Fe Co Ni Cu Zn

decreases remains same increases again

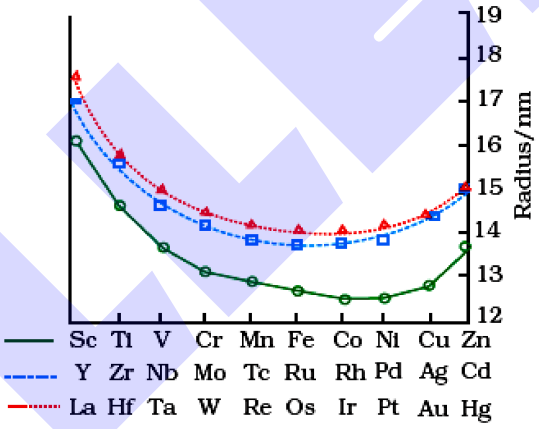


Fig. 8.3: Trends in atomic radii of transition elements

(v) Ionisation energy :

1st, 2nd, 3rd IE's are increasing from left to right for 1st Transition series, but not regularly.

For 2nd IE Cr > Fe > Mn and Cu > Zn

For 3rd IE Mn > Cr > Fe and Zn has highest.

(vi) DENSITY

(a) The atomic volume of the transition elements are low, compared with s-block, so their density is comparatively high (D = M/V)

(b) Os (22.57 gm cm^{-3}) and Ir (22.61 gm cm^{-3}) have highest density.

(c) In all the groups (except IIIB) there is normal increase in density from 3d to 4d series, and from 4d to 5d, it increases just double. Due to lanthanide contraction **Ex. :** Ti < Zr << Hf

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s-Soll And-Heating Effs.d-Block-Ron of Soll And-(OC)\Eng\04_d-Block\01_Th+Ex.p65

(d) **In 3d series**

	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Density/g cm^{-3}	3.43	4.1	6.07	7.19	7.21	7.8	8.7	8.9	8.9	7.1

(e) In 3d series highest density – Cu lowest density – Sc

(f) Some important orders of density

Fe < Ni < Cu Fe < Cu < Au Fe < Hg < Au

Electronic configurations and some other properties of the first series of transition elements

Element		Sc	Ti	v	Cr	Mn	Fe	Co	Ni	Cu	Zn
Atomic number		21	22	23	24	25	26	27	28	29	30
Electronic configuration											
	M	$3d^14s^2$	$3d^24s^2$	$3d^34s^2$	$3d^54s^2$	$3d^54s^2$	$3d^64s^2$	$3d^74s^2$	$3d^{10}4s^2$	$3d^{10}4s^1$	$3d^{10}4s^2$
	M^+	$3d^14s^1$	$3d^24s^1$	$3d^34s^1$	$3d^5$	$3d^54s^1$	$3d^64s^1$	$3d^74s^1$	$3d^84s^1$	$3d^{10}$	$3d^{10}4s^1$
	M^{2+}	$3d^1$	$3d^2$	$3d^3$	$3d^4$	$3d^5$	$3d^6$	$3d^7$	$3d^8$	$3d^9$	$3d^{10}$
	M^{3+}	[Ar]	$3d^1$	$3d^2$	$3d^3$	$3d^4$	$3d^5$	$3d^6$	$3d^7$	–	–
Enthalpy of atomisation, $\Delta_aH^\ominus$/kJ mol^{-1}		326	473	515	397	281	416	425	430	339	126
Ionisation Enthalpy, $\Delta_1H^\ominus$/kJ mol^{-1}											
$\Delta_1H^\ominus$	I	631	656	650	653	717	762	758	736	745	906
	II	1235	1309	1414	1592	1509	1561	1644	1752	1958	1734
	III	2393	2657	2833	2990	3260	2962	3243	3402	3556	3829
Metallic/Ionic	M	164	147	135	129	137	126	125	125	128	137
radii/pm	M^{2+}	–	–	79	82	82	77	74	70	73	75
	M^{3+}	73	67	64	62	65	65	61	60	–	–
Standard											
electrode	M^{2+}/M	–	–1.63	–1.18	–0.90	–1.18	–0.44	–0.28	–0.25	+0.34	–0.76
potential $E^\ominus$/V	M^{3+}/M^{2+}	–	–0.37	–0.26	–0.41	+1.57	+0.77	+1.97	–	–	–
Density/g cm^{-3}		3.43	4.1	6.07	7.19	7.21	7.8	8.7	8.9	8.9	7.1

VARIABLE OXIDATION STATES POSSIBLE :

(1) The elements which give the greatest number of oxidation states occur in or near the middle of the series. Manganese, for example, exhibits all the oxidation states from +2 to +7.

(2) The lesser number of oxidation states at the extreme ends stems from either too few electrons to lose or share (Sc, Ti) or too many d electrons (hence fewer orbitals available in which to share electrons with others) for higher valence (Cu, Zn).

(3) Thus, early in the series scandium(II) is virtually unknown and titanium (IV) is more stable than Ti(III) or Ti(II).

(4) At the other end, the only oxidation state of zinc is +2 (no d electrons are involved).

(5) The maximum oxidation states of reasonable stability correspond in value to the sum of the s and d electrons upto manganese ($Ti^{IV}O_2$, $V^{V}O_2^+$, $Cr^{VI}O_4^{2-}$, $Mn^{VII}O_4^-$) followed by a rather abrupt decrease in stability of higher oxidation states, so that the typical species to follow are $Fe^{II,III}$, $Co^{II,III}$, Ni^{II}, $Cu^{I,II}$, Zn^{II}.

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\6-Salt Anal+Heating Eff+s,d Block+Ion of Salt Anal+(OC)\Eng\04_d-Block\01_Th+Ex.p65

(6) The variability of oxidation states, a characteristic of transition elements, arises out of incomplete filling of d orbitals in such a way that their oxidation states differ from each other by unity, e.g., V^{II}, V^{III}, V^{IV}, V^{V}.

(7) This is in contrast with the variability of oxidation states of non transition elements where oxidation states normally differ by a unit of two.

(8) An interesting feature in the variability of oxidation states of the d–block elements is noticed among the groups (groups 4 through 10).

(9) In group 6, Mo(VI) and W(VI) are found to be more stable than Cr(VI). Thus Cr(VI) in the form of dichromate in acidic medium is a strong oxidising agent, whereas MoO_3 and WO_3 are not.

(10) Low oxidation states are found when a complex compound has ligands capable of π-acceptor character in addition to the σ-bonding. For example, in $Ni(CO)_4$ and $Fe(CO)_5$, the oxidation state of nickel and iron is zero.

(11) As the oxidation number of a metal increases, ionic character decreases. In the case of Mn, Mn_2O_7 is a covalent green oil. Even CrO_3 and V_2O_5 have low melting points. In these higher oxides, the acidic character is predominant. Thus, Mn_2O_7 gives $HMnO_4$ and CrO_3 gives H_2CrO_4 and $H_2Cr_2O_7$. V_2O_5 is, however, amphoteric though mainly acidic and it gives VO_4^{3-} as well as VO_2^+ salts. In vanadium there is gradual change from the basic V_2O_3 to less basic V_2O_4 and to amphoteric V_2O_5. V_2O_4 dissolves in acids to give VO^{2+} salts. Similarly, V_2O_5 reacts with alkalies as well as acids to give VO_4^{3-} and VO_2^+ respectively. The well characterised CrO is basic but Cr_2O_3 is amphoteric.

Oxidation states of the Ist transition series

most common ones are in bold types :

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
			+1					+1	
	+2	+2	+2	**+2**	**+2**	**+2**	**+2**	**+2**	**+2**
+3	**+3**	**+3**	**+3**	+3	**+3**	**+3**	+3		
	+4	+4	+4	**+4**	+4	+4	+4		
		+5	+5	+5					
			+6	+6	+6				
				+7					

Trends in stability of higher oxidation state :

(1) Table shows the stable halides of the 3*d* series of transition metals. The highest oxidation numbers are achieved in TiX_4 (tetrahalides), VF_5 and CrF_6. The +7 state for Mn is not represented in simple halides but MnO_3F is known, and beyond Mn no metal has a trihalide except FeX_3 and CoF_3.

(2) The ability of fluorine to stabilise the highest oxidation state is due to either higher lattice energy as in the case of CoF_3, or higher bond enthalpy terms for the higher covalent compounds, e.g., VF_5 and CrF_6.

(3) Although V^V is represented only by VF_5, the other halides, however, undergo hydrolysis to give oxohalides, VOX_3.

(4) Another feature of fluorides is their instability in the low oxidation states e.g., VX_2 (X = CI, Br or I)

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\6-Salt Anal-Heating Effs, d-Block-Ion of Salt Anal\(OC)\Eng\04_d-Block\01_Th+Ex.p65

Formulas of halides of 3d-metals

Oxidation Number									
+ 6			CrF_6						
+ 5		VF_5	CrF_5						
+ 4	TiX_4	VX_4^I	CrX_4	MnF_4					
+ 3	TiX_3	VX_3	CrX_3	MnF_3	FeX_3^I	CoF_3			
+ 2	TiX_2^{III}	VX_2	CrX_2	MnX_2	FeX_2	CoX_2	NiX_2	CuX_2^{II}	ZnX_2
+ 1								CuX^{III}	

Key : $X = F \rightarrow I$; $X^I = F \rightarrow Br$; $X^{II} = F \rightarrow Cl$; $X^{III} = Cl \rightarrow I$

and the same applies to CuX. On the other hand, all Cu(II) halides are known except the iodide. In this case, Cu^{2+} oxidises I^- to I_2:

$$2Cu^{2+} + 4I^- \rightarrow Cu_2I_2(s) + I_2$$

(5) However, many copper (I) compounds are unstable in aqueous solution and undergo disproportionation.

$$2Cu^+ \rightarrow Cu^{2+} + Cu$$

(6) The stability of $Cu^{2+}(aq)$ rather than $Cu^+(aq)$ is due to the much more negative $\Delta_{Hyd}H^\ominus$ of $Cu^{2+}(aq)$ than Cu^+, which more than compensates for the second ionisation enthalpy of Cu.

(7) The ability of oxygen to stabilise the highest oxidation state is demonstrated in the oxides.

(8) The highest oxidation number in the oxides coincides with the group number and is attained in Sc_2O_3 to Mn_2O_7.

(9) Beyond Group 7, no higher oxides of Fe above Fe_2O_3, are known, although ferrates (VI)$(FeO_4)^{2-}$, are formed in alkaline media but they readily decompose to Fe_2O_3 and O_2.

(10) Besides the oxides, oxocations stabilise V^v as VO_2^+, V^{IV} as VO^{2+} and Ti^{IV} as TiO^{2+}

(11) The ability of oxygen to stabilise these high oxidation states exceeds that of fluorine. Thus the highest Mn fluoride is MnF_4 whereas the highest oxide is Mn_2O_7. The ability of oxygen to form multiple bonds to metals explains its superiority.

(12) In the covalent oxide Mn_2O_7, each Mn is tetrahedrally surrounded by O's including a Mn–O–Mn bridge.

(13) The tetrahedral $[MO_4]^n$ ions are known for V^V, Cr^{VI}, Mn^V, Mn^{VI} and Mn^{VII}.

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\d-Sub Ano+Heating Eff+s,d-Block+Ion of Salt Ano+(OC)\Eng\04_d-Block\01_Th+Ex.p65

FORMATION OF COLOURED ION :

Colour : (aquated)

Ti^{4+} ⟶ colourless

V^{4+} ⟶ blue

V^{2+} ⟶ violet

Cr^{3+} ⟶ violet

Mn^{2+} ⟶ light pink

Fe^{3+} ⟶ yellow

Ni^{2+} ⟶ green

Zn^{2+} ⟶ colourless

Sc^{3+} ⟶ colourless

Ti^{3+} ⟶ purple

V^{3+} ⟶ green

Cr^{2+} ⟶ blue

Mn^{3+} ⟶ violet

Fe^{2+} ⟶ light green

Co^{2+} ⟶ pink

Cu^{2+} ⟶ blue

CATALYTIC PROPERTIES

(1) The transition metals and their compounds are known for their catalytic activity. This activity is ascribed to their ability to adopt multiple oxidation states and to form complexes. Vanadium(V) oxide (in Contact Process), finely divided iron (in Haber's Process), and nickel (in Catalytic Hydrogenation) are some of the examples.

(2) Catalysts at a solid surface involve the formation of bonds between reactant molecules and atoms of the surface of the catalyst (first row transition metals utilise 3d and 4s electrons for bonding).

(3) This has the effect of increasing the concentration of the reactants at the catalyst surface and also weakening of the bonds in the reacting molecules (the activation energy is lowering).

(4) Also because the transition metal ions can change their oxidation states, they become more effective as catalysts. For example, iron(III) catalyses the reaction between iodide and persulphate ions.

$$2I^- + S_2O_8^{2-} \rightarrow I_2 + 2SO_4^{2-}$$

An explanation of this catalytic action can be given as:

$$2Fe^{3+} + 2I^- \rightarrow 2\ Fe^{2+} + I_2$$

$$2\ Fe^{2+} + S_2O_8^{2-} \rightarrow 2Fe^{3+} + 2SO_4^{2-}$$

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s-Salt Anal-Heating Effs,d-Block-Ron of Salt Anal(OC)\Eng\04_d-Block\01_Th+Ex.p65

Catalyst		Used
$TiCl_3$	⟶	Used as the Ziegler-Natta catalyst in the production of polythene.
V_2O_5	⟶	Convert SO_2 to SO_3 in the contact process for making H_2SO_4
MnO_2	⟶	Used as a catalyst to decompose $KClO_3$ to give O_2
Fe	⟶	Promoted iron is used in the Haber-Bosch process for making NH_3
$FeCl_3$	⟶	Used in the production of CCl_4 from CS_2 and Cl_2
$PdCl_2$	⟶	Wacker process for converting $C_2H_4 + H_2O + PdCl_2$ to $CH_3CHO + 2HCl + Pd$.
Pd	⟶	Used for hydrogenation (e.g. phenol to cyclohexanone).
Pt/PtO	⟶	Adams catalyst, used for reductions.
Pt	⟶	Formerly used for $SO_2 \longrightarrow SO_3$ in in the contace process for making H_2SO_4
Pt/Rh	⟶	Formerly used in the ostwald process for making HNO_3 to oxidize NH_3 to NO
Cu	⟶	Is used in the direct process for manufacture of $(CH_3)_2SiCl_2$ used to make silicones.
Cu/V	⟶	Oxidation of cyclohexanol/cyclohexanone mixture to adipic acid which is used to make nylon-66
$CuCl_2$	⟶	Deacon process of making Cl_2 from HCl
Ni	⟶	Raney nickel, numerous reduction processes (e.g. manufacture of hexamethylenediamine, productiomn of H_2 from NH_3, reducing anthraquinone to anthraquinol in the production of H_2O_2
$FeSO_4 + H_2O_2$	⟶	Used as Fenton's reagent for oxidizing alcohols to aldehydes.

Formation of Interstitial Compounds

Interstitial compounds are those which are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals. The principal physical and chemical characteristics of these compounds are as follows:

(i) They have high melting points, higher than those of pure metals.

(ii) They are very hard, some borides approach diamond in hardness.

(iii) They retain metallic conductivity.

(iv) They are chemically inert.

Alloy Formation

An alloy is a blend of metals prepared by mixing the components. Alloys may be homogeneous solid solutions in which the atoms of one metal are distributed randomly among the atoms of the other. Such alloys are formed by atoms with metallic radii that are within about 15 percent of each other. Because of similar radii and other characteristics of transition metals, alloys are readily formed by these metals. The alloys so formed are hard and have often high melting points.

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\d-Sub Ano-Heating Eff+s_d-Block-Run of Salt Ano-(OC)\Eng\04_d-Block\01_Th+Ex.p65

The best known are ferrous alloys: chromium, vanadium, tungsten, molybdenum and manganese are used for the production of a variety of steels and stainless steel. Alloys of transition metals with non transition metals such as brass (copper-zinc) and bronze (copper-tin), are also of considerable industrial importance.

CHROMATE -DICHROMATE

Preparation : $4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \xrightarrow[\text{red hot in presence of air}]{1000°-1300°C} \underbrace{8Na_2CrO_4 + 2Fe_2O_3}_{\text{Residue}(Fe_2O_3)} + 8CO_2\uparrow$

Filtrate(Na_2CrO_4) — Dissolve in water and filtered

(chromite ore)

[Lime (CaO) added with Na_2CO_3 which keeps the mass porous so that air has access to all parts and prevents fusion]

Then, $2Na_2CrO_4 + H_2SO_4 \longrightarrow Na_2SO_4\downarrow + Na_2Cr_2O_7 + H_2O$

conc. It's solubility upto 32°C increases and then decreases } Hence, suitable temp. is to be employed to crystallise out Na_2SO_4 first.

Then $Na_2Cr_2O_7$ is crystallised out as $Na_2Cr_2O_7 \cdot 2H_2O$ on evaporation.

(red crystal)

How to get $K_2Cr_2O_7$:

$$\underset{\text{hot conc.}}{Na_2Cr_2O_7} + 2KCl \xrightarrow[\text{decomposition}]{\text{double}} K_2Cr_2O_7 + 2NaCl$$

NaCl crystallises out first and filtered off. Then $K_2Cr_2O_7$ crystallised out on cooling

The chromates and dichromates are interconvertible in aqueous solution depending upon pH of the solution. The oxidation state of chromium in chromate and dichromate is the same.

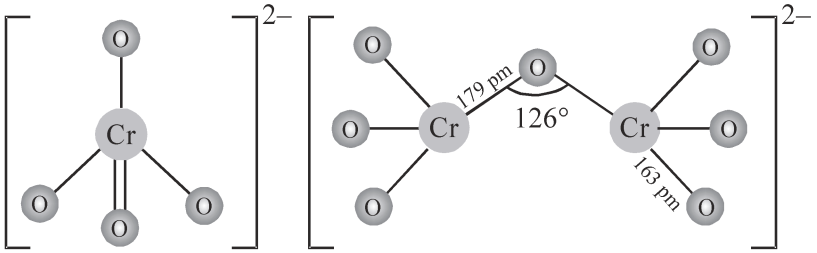
$$2\ CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$$

$$Cr_2O_7^{2-} + 2OH^- \rightarrow 2CrO_4^{2-} + H_2O$$

The structures of chromate ion, CrO_4^{2-} and the dichromate ion, $Cr_2O_7^{2-}$ are shown below. The chromate ion is tetrahedral whereas the dichromate ion consists of two tetrahedra sharing one corner with Cr–O–Cr bond angle of 126°. Sodium and potassium dichromates are strong oxidising agents; the sodium salt has a greater solubility in water and is extensively used as an oxidising agent in organic chemistry. Potassium dichromate is used as a primary standard in volumetric analysis. In acidic solution, its oxidising action can be represented as follows:

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O\ (E^\ominus = 1.33V)$$

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\6-Salt Anal-Heating Effs.,d-Block-Ron of Salt Anal-(OC)\Eng\04_d-Block\01_Th+Ex.p65



Chromate ion Dichromate ion

Thus, acidified potassium dichromate will oxidise iodides to iodine, sulphides to sulphur, tin(II) to tin(IV) and iron(II) salts to iron(III). The half-reactions are noted below:

$6\ I^- \rightarrow 3I_2 + 6e^-$; $3\ Sn^{2+} \rightarrow 3Sn^{4+} + 6e^-$

$3H_2S \rightarrow 6H^+ + 3S + 6e^-$; $6\ Fe^{2+} \rightarrow 6Fe^{3+} + 6e^-$

The full ionic equation may be obtained by adding the half-reaction for potassium dichromate to the half-reaction for the reducing agent, for e.g.,

$$Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$$

* Similarities between hexavalent Cr & S-compounds :

(i) SO_3 & CrO_3 $\longrightarrow$ both acidic.

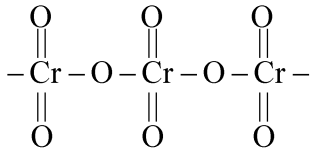
(ii) S $\longrightarrow$ SO_4^{2-} , $S_2O_7^{2-}$, Cr $\longrightarrow$ CrO_4^{2-} , $Cr_2O_7^{2-}$

(iii) CrO_4^{-2} & SO_4^{2-} are isomorphous

(iv) SO_2Cl_2 & CrO_2Cl_2 $\xrightarrow{OH^-}$ SO_4^{2-} & CrO_4^{2-} respectively.

(v) SO_3Cl^- & CrO_3Cl^- $\xrightarrow{OH^-}$ SO_4^{2-} & CrO_4^{2-}

(vi) CrO_3 & $\beta(SO_3)$ has same structure



Q. In laboratory $K_2Cr_2O_7$ is used mainly not $Na_2Cr_2O_7$. Why?

Sol. $Na_2Cr_2O_7$ is deliquescent enough and changes its concentration and can not be taken as primary standard solution whereas $K_2Cr_2O_7$ has no water of crystallisation and not deliquescent.

Q. How to standardise $Na_2S_2O_3$ solution in iodometry?

Sol. $K_2Cr_2O_7$ is primary standard $\Rightarrow$ strength is known by weighing the salt in chemical balance and dissolving in measured amount of water.

Then in acidic solution add. KI

$$Cr_2O_7^{2-} + 14H^+ + 6I^- \longrightarrow 2Cr^{3+} + 3I_2 + 7H_2O$$

This I_2 is liberated can be estimated with $S_2O_3^{2-}$.

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\d-Sall Ana+Heating Eff+s_d-Block+Ion of Sall Ana+(OC)\Eng\04_d-Block\01_Th+Ex.p65

MANGANATE & PERMANGANATE

PREPARATION OF MANGANATE (MnO_4^{2-}) :-

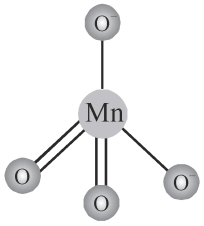
$MnO_2 \xrightarrow[\text{or NaOH, } \Delta \text{ in presence of air}]{KOH} \underset{\text{green melt}}{MnO_4^{2-}}$ $[2MnO_2 + 4KOH + O_2 \longrightarrow 2K_2MnO_4 + 2H_2O]$

$MnO_2 + 2KOH + KNO_3 \longrightarrow K_2MnO_4 + KNO_2 + H_2O$

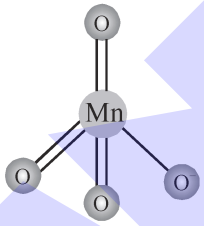
$3MnO_2 + 6KOH + KClO_3 \longrightarrow 3K_2MnO_4 + KCl + 3H_2O$

Cold Water + little alkali → Green soln. → On evaporation → If KOH used → K_2MnO_4 (isomorphrous with K_2SO_4)

On evaporation → If NaOH used → $Na_2MnO_4.10H_2O$ isomorphrous with $Na_2SO_4.10H_2O$



Tetrahedral manganate (green) ion



Tetrahedral permanganate (purple) ion

In presence of $KClO_3$ & KNO_3 the above reaction is more faster because these two on decomposition provides O_2 easily.

* Manganate is also obtained when $KMnO_4$ is boiled with KOH.

$4KMnO_4 + 4KOH \xrightarrow{\text{boiled}} 4K_2MnO_4 + 2H_2O + O_2$

Properties : The above green solution is quite stable in alkali, but in pure water and in presence of acids, depositing MnO_2 and giving a purple solution of permanganate.

$3K_2MnO_4 + 2H_2O \longrightarrow \underset{\text{purple}}{2KMnO_4} + \underset{\text{drak brown}}{MnO_2\downarrow} + 4KOH$

Prob. : $E^o_{MnO_4^{2-}/MnO_2} = 2.26$ V ; $E^o_{MnO_4^{2-}/MnO_4^-} = -0.56$ V

Prove that MnO_4^{2-} will disproportionate in acidic medium.

Conversion of MnO_4^{2-} to MnO_4^-

$3K_2MnO_4 + 2H_2SO_4 \longrightarrow 2KMnO_4 + MnO_2\downarrow + 2K_2SO_4 + 2H_2O$

or $3K_2MnO_4 + 2H_2O + 4CO_2 \longrightarrow 2KMnO_4 + MnO_2 + 4KHCO_3$

But in the above method $\frac{1}{3}$ of Mn is lost as MnO_2 but when oxidised either by Cl_2 or by O_3

$2K_2MnO_4 + Cl_2 \longrightarrow 2KMnO_4 + 2KCl$ [Unwanted MnO_2 does not form]

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s-Soll And-Heating Effs.d-Block-Ron of Soll And-(OC)\Eng\04_d-Block\01_Th+Ex.p65

OR

$$2K_2MnO_4 + O_3 + H_2O \longrightarrow 2KMnO_4 + 2KOH + O_2$$

OR

$$\underset{\text{Manganate}}{MnO_4^{2-}} \xrightarrow[\text{alkaline solution}]{\text{Electrolytic oxidation in}} \underset{\text{permanganate ion}}{MnO_4^-}$$

Oxidising Prop. of $KMnO_4$: (in acidic medium)

(i) $MnO_4^- + Fe^{+2} + H^+ \longrightarrow Fe^{+3} + Mn^{+2} + H_2O$

(ii) $MnO_4^- + I^- + H^+ \longrightarrow Mn^{+2} + I_2 + H_2O$

(iii) $MnO_4^- + H_2O_2 + H^+ \longrightarrow Mn^{+2} + O_2 + H_2O$

(iv) $MnO_4^- + SO_2 \xrightarrow{H^\oplus} Mn^{+2} + H_2SO_4$

(v) $MnO_4^- + NO_2^- + H^+ \longrightarrow Mn^{+2} + NO_3^- + H_2O$

(vi) $MnO_4^- + H_2C_2O_4 + H^+ \longrightarrow Mn^{+2} + CO_2 + H_2O$

(vii) $MnO_4^- + H_2S \longrightarrow Mn^{2+} + S\downarrow + H_2O$

(viii) $MnO_4^- + S_2O_3^{2-} \longrightarrow Mn^{2+} + S\downarrow + SO_4^{2-}$

(1)* It is not a primary standard since it is difficult to get it in a high degree of purity and free from traces of MnO_2.

(2)* It is slowly reduced to MnO_2 especially in presence of light or acid

$$4MnO_4^- + 4H^+ \longrightarrow 4MnO_2 + 2H_2O + 3O_2$$

Hence it should be kept in dark bottles and standardise just before use.

(ix) $2KMnO_4 + 16HCl \longrightarrow 2KCl + 5Cl_2 + 8H_2O + 2MnCl_2$

Note*: *Permanganate titrations in presence of hydrochloric acid are unsatisfactory since hydrochloric acid is oxidised to chlorine.

Oxidising Prop. of $KMnO_4$ in neutral or faintly alkaline solution.

$2MnO_4^- + 2OH^- \longrightarrow 2MnO_4^{2-} + H_2O + O$. Then $2MnO_4^{2-} + 2H_2O \longrightarrow 2MnO_2 + 4OH^- + 2O$

(i) $2KMnO_4 + H_2O + KI \longrightarrow 2MnO_2 + 2KOH + KIO_3$

(ii) $2KMnO_4 + 3HCO_2K \longrightarrow 2MnO_2 + KHCO_3 + 2K_2CO_3 + H_2O$

(iii) $2KMnO_4 + 3H_2O_2 \longrightarrow 2KOH + 2MnO_2 + 2H_2O + 3O_2$

(iv) Thiosulphate is oxidised almost quantitatively to sulphate:

$$8MnO_4^- + 3S_2O_3^{2-} + H_2O \longrightarrow 8MnO_2 + 6SO_4^{2-} + 2OH^-$$

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\d-Sub Ano-Heating Eff+s,d-Block-Ron of Salt Ano-(OC)\Eng\04_d-Block\01_Th+Ex.p65

Oxidising Prop. in neutral or weakly acidic solution:

(i) $2KMnO_4 + 3MnSO_4 + 2H_2O \xrightarrow{\text{in presence } Zn^{+2} \text{ or } ZnO} 5MnO_2 + K_2SO_4 + 2H_2SO_4$

or $MnO_4^- + Mn^{+2} + 2H_2O \longrightarrow 5MnO_2 + 4H^+$

In absence of Zn^{+2} ions, some of the Mn^{+2} ion may escape, oxidation through the formation of insoluble $Mn^{II}[Mn^{IV}O_3]$ manganous permanganite.

Uses of $KMnO_4$:

Besides its use in analytical chemistry, potassium permanganate is used as a favourite oxidant in preparative organic chemistry. Its uses for the bleaching of wool, cotton, silk and other textile fibres and for the decolourisation of oils are also dependent on its strong oxidising power.

** In laboratory conversion of Mn^{+2} to MnO_4^- is done by :

(i) PbO_2 (ii) $Pb_3O_4 + HNO_3$ (iii) $Pb_2O_3 + HNO_3$ (iv) $NaBiO_3 / H^+$
(v) $(NH_4)_2S_2O_8 / H^+$ (vi) KIO_4 / H^+

Heating effect : $2KMnO_4 \xrightarrow[200°C]{\Delta} K_2MnO_4 + MnO_2 + O_2$

green Black

$2K_2MnO_4 \xrightarrow[\text{hot}]{\text{at red}} 2K_2MnO_3 + O_2$

SILVER AND ITS COMPOUND

(I) Metallic Ag

- aq. regia → Not dissolved
- Dry or Moist air → No effect
- H_2S → Black $[4Ag + 2H_2S + O_2 \longrightarrow 2Ag_2S + 2H_2O]$
- H_2SO_4 → $2Ag + 2H_2SO_4 \longrightarrow Ag_2SO_4 + SO_2 + H_2O$
- HNO_3 hot conc. → $Ag + 2HNO_3 \longrightarrow AgNO_3 + NO_2 + H_2O$
- HCl → Not affected dil. HCl

$E^o_{Ag/Ag^+} < E^o_{H_2/H^+}$

Hence in presence of O_2.
Ag reacts with dil.HCl
$4Ag + 4HCl + O_2 \longrightarrow 4AgCl\downarrow + 2H_2O$

In the same way in presence of O_2, Ag complexes with NaCN / KCN.

$4Ag + 8KCN + 2H_2O + O_2 \longrightarrow 4K[Ag(CN)_2] + 4KOH$

$AgNO_3$

Prepration : Reaction of Ag with dilute HNO_3 or conc. HNO_3.

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s-Soll Anal-Heating Effs,d-Block-Ron of Soll Anal-(OC)\Eng\04_d-Block\01_Th+Ex.p65

Properties :

(i) It is called as lunar caustic because in contact with skin it produces burning sensation like that of caustic soda with the formation of finely devided silver (black colour)

(ii) Thermal decomposition : $2AgNO_3(s) \rightarrow 2Ag\,(s) + O_2\,(g) + 2NO_2\,(g)$

(iii) Props. of $AgNO_3$

$6AgNO_3 + 3I_2 + 3H_2O \longrightarrow 5AgI + AgIO_3 + 6HNO_3$

(excess)

(iv) $Ag_2SO_4 \xrightarrow{\Delta} 2Ag + SO_2 + O_2$

(v) $A(AgNO_3) \xrightarrow[\text{added}]{B}$ white ppt appears quickly

$B(Na_2S_2O_3) \xrightarrow[\text{added}]{A}$ It takes time to give white ppt. } Explain

(vi) $Ag_2S_2O_3 + H_2O \xrightarrow{\Delta} Ag_2S + H_2SO_4$

AgCl, AgBr, AgI (but not Ag_2S) are soluble in $Na_2S_2O_3$ forming $[Ag(S_2O_3)_2]^{-3}$ complexes

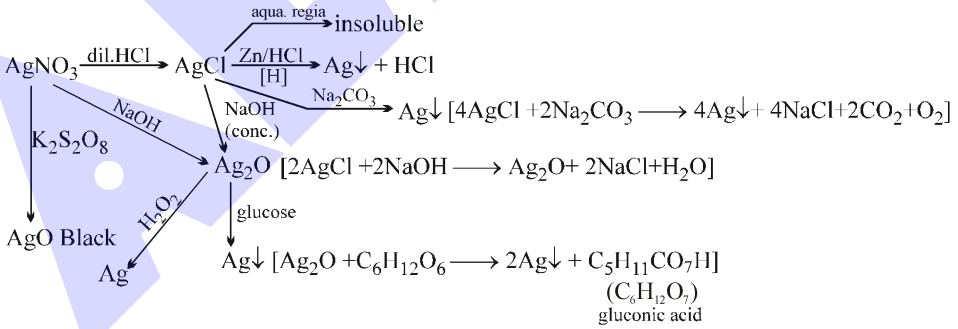
(vii) $AgBr + AgNO_3 \xrightarrow{KBr} AgBr\downarrow + KNO_3$

Pale yellow ppt.

Heating effect : $2AgNO_3 \xrightarrow{212°C} 2AgNO_2 + O_2$

$2AgNO_2 \xrightarrow{500°C} 2Ag + 2NO + O_2$

(viii)



$Ag_2O + H_2O_2 \longrightarrow 2Ag + H_2O + O_2$

$K_2S_2O_8 + 2AgNO_3 + 2H_2O \longrightarrow 2AgO + 2KHSO_4 + 2HNO_3$

* AgO supposed to be paramagnetic due to d^9 configuration. But actually it is diamagnetic and exists as $Ag^I\,[Ag^{III}O_2]$

* Reaction involved in developer :

$K_2Fe^{II}(C_2O_4)_2 + AgBr \longrightarrow KFe^{III}\,(C_2O_4)_2 + Ag\downarrow + KBr$

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\d-Salt Ana+Heating Eff+s,d-Block+Ion of Salt Ana+(IOC)\Eng\04_d-Block\01_Th+Ex.p65

ZINC COMPOUNDS

ZnO

It is called as phillospher's wool due to its wooly flock type appearance

Preparation: [1] $2Zn + O_2 \longrightarrow 2ZnO$

[2] Calcination of $ZnCO_3$ or $Zn(NO_3)_2$ or $Zn(OH)_2$

Purest ZnO : $4ZnSO_4 + 4Na_2CO_3 + 3H_2O \longrightarrow \underset{\text{white basic zinc carbonate}}{ZnCO_3 \cdot 3Zn(OH)_2} \downarrow + 4Na_2SO_4 + 3CO_2$

$\downarrow \Delta$

$\underset{\text{pure}}{4ZnO} + 3H_2O \uparrow + CO_2 \uparrow$

Properties: 1] $ZnO(\text{cold}) \underset{}{\overset{\Delta}{\rightleftharpoons}} ZnO(\text{hot})$

white yellow

2] It is insoluble in water

3] It sublimes at 400°C

4] It is amphoteric oxide , react with acid & base both.

$ZnO + 2HCl \longrightarrow ZnCl_2 + H_2O$

$ZnO + H_2SO_4 \longrightarrow ZnSO_4 + H_2O$

$ZnO + 2NaOH \longrightarrow Na_2ZnO_2 + H_2O$

5] $ZnO \longrightarrow Zn$ by H_2 & C

$ZnO + H_2 \xrightarrow{>400°C} Zn + H_2O$

$ZnO + C \longrightarrow Zn + CO$

6] It forms Rinmann's green with $Co(NO_3)_2$

$2Co(NO_3)_2 \longrightarrow 2CoO + 4NO_2 + O_2$

$CoO + ZnO \longrightarrow \underset{\text{Rinmann's green}}{CoZnO_2} \text{ or } CoO \cdot ZnO$

Uses: (1) As white pigment, it is superior than white lead because it does not turn into black

(2) Rinmann's green is used as green pigment

(3) It is used as zinc ointment in medicine

$ZnCl_2$

Preparation : $ZnO + 2HCl \longrightarrow ZnCl_2 + H_2O$

$ZnCO_3 + 2HCl \longrightarrow ZnCl_2 + H_2O + CO_2$ } It crystallises as $ZnCl_2 \cdot 2H_2O$

$Zn(OH)_2 + 2HCl \longrightarrow ZnCl_2 + 2H_2O$

Anh. $ZnCl_2$ cannot be made by heating $ZnCl_2 \cdot 2H_2O$ because

$ZnCl_2 \cdot 2H_2O \xrightarrow{\Delta} Zn(OH)Cl + HCl + H_2O$

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s-Salt Anal-Heating Effs,d-Block-Ron of Salt Anal(OC)\Eng\04_d-Block\01_Th+Ex.p65

$Zn(OH)Cl \xrightarrow{\Delta} ZnO + HCl$

To get anh. $ZnCl_2$: $Zn + Cl_2 \longrightarrow ZnCl_2$

$Zn + 2HCl(dry) \longrightarrow ZnCl_2 + H_2$

or $Zn + HgCl_2 \longrightarrow ZnCl_2 + Hg$

Properties: (i) It is deliquescent white solid (when anhydrous)

(ii) $ZnCl_2 + H_2S \longrightarrow ZnS$

" $+ NaOH \longrightarrow Zn(OH)_2 \xrightarrow{\text{excess}} Na_2[Zn(OH)_4]$

" $+ NH_4OH \longrightarrow Zn(OH)_2 \xrightarrow{\text{excess}} [Zn(NH_3)_4]^{2+}$

Uses: 1] Used for impregnating timber to prevent destruction by insects

2] As dehydrating agent when anhydrous

3] $ZnO \cdot ZnCl_2$ used in dental filling

$ZnSO_4$

Preparation:→

$Zn + \text{dil } H_2SO_4 \longrightarrow ZnSO_4 + H_2$

$ZnO + \text{dil } H_2SO_4 \longrightarrow ZnSO_4 + H_2O$

$ZnCO_3 + \text{dil } H_2SO_4 \longrightarrow ZnSO_4 + H_2O + CO_2$

$ZnS + 2O_2 \longrightarrow ZnSO_4$

$ZnS + \frac{3}{2}O_2 \longrightarrow ZnO + SO_2$ } parallel reaction

$ZnS + 4O_3 \longrightarrow ZnSO_4 + 4O_2$

Props.1] $ZnSO_4 \cdot 7H_2O \xrightarrow{39-70^\circ C} ZnSO_4 \cdot 6H_2O \xrightarrow{>70^\circ C} ZnSO_4 \cdot H_2O \xrightarrow{>280^\circ C} ZnSO_4$

$\frac{1}{2}O_2 + SO_2 + ZnO \xleftarrow{>800^\circ C}$

Uses: 1] in eye lotion

2] Lithophone ($ZnS + BaSO_4$) is used as white pigment

COPPER COMPOUNDS

CuO

Preparation : (i) $CuCO_3.Cu(OH)_2 \xrightarrow{\Delta} 2CuO + H_2O + CO_2$ (Commercial process)

Malachite Green

(native Cu-carbonate)

(ii) $2Cu + O_2 \longrightarrow 2CuO$ & $Cu_2O + \frac{1}{2}O_2 \longrightarrow 2CuO$

(iii) $Cu(OH)_2 \xrightarrow{\Delta} CuO + H_2O$

(iv) $2Cu(NO_3)_2 \xrightarrow{250^\circ C} 2CuO + 4NO_2 + O_2$

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\d-Sath Ana+Heating Eff+s_d-Block+Ion of Salt Ana+(OC)\Eng\04_d-Block\01_Th+Ex.p65

Properties : (i) CuO is insoluble in water

(ii) Readily dissolves in dil. acids

$CuO + H_2SO_4 \longrightarrow CuSO_4 + H_2O$

$HCl \longrightarrow CuCl_2$

$HNO_3 \longrightarrow Cu(NO_3)_2$

(iii) It decomposes when, heated above 1100°C

$4CuO \longrightarrow 2Cu_2O + O_2$

(iv) CuO is reduced to Cu by H_2 or C under hot condition

$CuO + C \longrightarrow Cu + CO \uparrow$

$CuO + H_2 \longrightarrow Cu + H_2O \uparrow$

$CuCl_2$

Preparation:→ $CuO + 2HCl(conc.) \longrightarrow CuCl_2 + H_2O$

$Cu(OH)_2 \cdot CuCO_3 + 4HCl \longrightarrow 2CuCl_2 + 3H_2O + CO_2$

Properties :→ (i) It is crystallised as $CuCl_2 \cdot 2H_2O$ of Emerald green colour

(ii) dil. solution in water is blue in colour due to formation of $[Cu(H_2O)_4]^{2+}$complex.

(iii) When conc. HCl or KCl added to dil. solution of $CuCl_2$ the colour changes into yellow, owing to the formation of $[CuCl_4]^{2-}$

(iv) The conc. aq. solution is green in colour having the two complex ions in equilibrium $2[Cu(H_2O)_4]Cl_2 \rightleftharpoons [Cu(H_2O)_4]^{2+} + [CuCl_4]^{2-} + 4H_2O$

(v) $CuCl_2 \longrightarrow CuCl$ by no. of reagents

(a) $CuCl_2 + Cu\text{-turnings} \xrightarrow{\Delta} 2CuCl$

(b) $2CuCl_2 + H_2SO_3 + H_2O \longrightarrow 2CuCl + 2HCl + 2H_2SO_4$

(c) $2CuCl_2 + Zn/HCl \longrightarrow 2CuCl + ZnCl_2$

(d) $CuCl_2 + SnCl_2 \longrightarrow CuCl + SnCl_4$

** $CuF_2 \cdot 2H_2O \longrightarrow$ light blue

$CuCl_2 \cdot 2H_2O \longrightarrow$ green

$CuBr_2 \longrightarrow$ almost black

} Anh. $CuCl_2$ is dark brown mass obtained by heating $CuCl_2 \cdot 2H_2O$ at 150°C in presence of HCl vap.

CuI_2 does not exist $\quad CuCl_2 \cdot 2H_2O \xrightarrow[HCl\ gas]{150°C} CuCl_2 + 2H_2O$

$CuSO_4$

Preparation :→ $CuO + H_2SO_4(dil) \longrightarrow CuSO_4 + H_2O$

$Cu(OH)_2 + H_2SO_4(dil) \longrightarrow CuSO_4 + 2H_2O$

$Cu(OH)_2 \cdot CuCO_3 + H_2SO_4\ (dil) \rightarrow CuSO_4 + 3H_2O + CO_2$

$Cu + H_2SO_4 + \frac{1}{2}O_2 \longrightarrow CuSO_4 + H_2O$ [Commercial scale]

(Scrap)

$Cu + dil.\ H_2SO_4 \longrightarrow$ no reaction {Cu is a below H in electrochemical series}

ALLEN®

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s-Salt Anal-Heating Effs.,d-Block-Ron of Salt Anal-(OC)\Eng\04_d-Block\01_Th+Ex.p65

Properties :→(i) It is crystallised as $CuSO_4 \cdot 5H_2O$

(ii) $CuSO_4 \cdot 5H_2O \underset{\text{effloresence}}{\overset{\text{On exposure}}{\rightleftharpoons}} CuSO_4 \cdot 3H_2O \overset{100°C}{\rightleftharpoons} CuSO_4 \cdot H_2O$

Blue take places Pale blue Bluish white

$\overset{230°C}{\rightleftharpoons}$

$CuSO_4$(anh.)
white

$\swarrow^{800°C}$ $CuO + SO_2 + \frac{1}{2}O_2$ | $\downarrow_{750°C}$ $CuO + SO_3$

(iii) Revision with all others reagent

IRON COMPOUNDS

$FeSO_4 \cdot 7H_2O$

Preparation:→(i) Scrap Fe + $H_2SO_4 \longrightarrow FeSO_4 + H_2\uparrow$
(dil.)

(ii) From Kipp's waste

$FeS + H_2SO_4(dil) \longrightarrow FeSO_4 + H_2S\uparrow$

(iii) $FeS_2 + 2H_2O + \frac{7}{2}O_2 \longrightarrow FeSO_4 + H_2SO_4$

Properties:→ (i) It undergoes aerial oxidation forming basic ferric sulphate

$4FeSO_4 + H_2O + O_2 \longrightarrow 4Fe(OH)SO_4$

(ii) $FeSO_4 \cdot 7H_2O \xrightarrow{300°C} \underset{\text{anh. white}}{FeSO_4} \xrightarrow[\text{temp.}]{\text{high}} Fe_2O_3 + SO_2 + SO_3$

(iii) Aq. solution is acidic due to hydrolysis

$FeSO_4 + 2H_2O \rightleftharpoons Fe(OH)_2 + H_2SO_4$
weak base

(iv) It is a reducing agent

(a) $Fe^{2+} + MnO_4^- + H^+ \longrightarrow Fe^{3+} + Mn^{2+} + H_2O$

(b) $Fe^{2+} + Cr_2O_7^{2-} + H^+ \longrightarrow Fe^{3+} + Cr^{3+} + H_2O$

(c) $Au^{3+} + Fe^{2+} \longrightarrow Au + Fe^{3+}$

(d) $Fe^{2+} + HgCl_2 \longrightarrow Hg_2Cl_2\downarrow + Fe^{3+}$
white ppt.

(v) It forms double salt. Example $(NH_4)_2SO_4 \cdot FeSO_4 \cdot 6H_2O$

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\6-Salt Ana+Heating Eff+s,d-Block+Ion of Salt Ana+(OC)\Eng\04_d-Block\01_Th+Ex.p65

FeO(Black)

Preparation : $FeC_2O_4 \xrightarrow[\text{in absence of air}]{\Delta} FeO + CO + CO_2$

Properties : It is stable at high temperature and on cooling slowly disproportionates into Fe_3O_4 and iron

$$4FeO \longrightarrow Fe_3O_4 + Fe$$

$FeCl_2$

Preparation: $Fe + 2HCl \xrightarrow[\text{a current of HCl}]{\text{heated in}} FeCl_2 + H_2$

OR

$$2FeCl_3 + H_2 \xrightarrow{\Delta} 2FeCl_2 + 2HCl$$

Properties:→ (i) It is deliquescent in air like $FeCl_3$

(ii) It is soluble in water, alcohol and ether also because it is sufficiently covalent in nature

(iii) It volatilises at about 1000°C and vapour density indicates the presence of Fe_2Cl_4. Above 1300°C density becomes normal

(iv) It oxidises on heating in air

$$12FeCl_2 + 3O_2 \longrightarrow 2Fe_2O_3 + 8FeCl_3$$

(v) H_2 evolves on heating in steam

$$3FeCl_2 + 4H_2O \longrightarrow Fe_3O_4 + 6HCl + H_2$$

(vi) It can exist as different hydrated form

$FeCl_2 \cdot 2H_2O \longrightarrow$ colourless

$FeCl_2 \cdot 4H_2O \longrightarrow$ pale green

$FeCl_2 \cdot 6H_2O \longrightarrow$ green

Ferric Chloride ($FeCl_3$)

This is the most important ferric salt. It is known in anhydrous and hydrated forms. The hydrated form consists of six water molecules, $FeCl_3.6H_2O$.

Preparation :

(i) Anhydrous ferric chloride is obtained by passing dry chlorine gas over heated iron fillings. The vapours are condensed in a bottle attached to the outlet of the tube.

$$2Fe + 3Cl_2 \rightarrow 2FeCl_3$$

(ii) $Fe_2(CO_3)_3 + 6HCl \rightarrow 2FeCl_3 + 3H_2O + 3CO_2$

$Fe(OH)_3 + 3HCl \rightarrow FeCl_3 + 3H_2O$

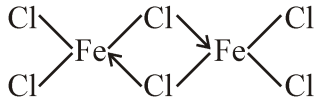
$Fe_2O_3 + 6HCl \rightarrow 2FeCl_3 + 3H_2O$

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\d-Salt Anal-Heating Effs,d-Block-Ron of Salt Anal(OC)\Eng\04_d-Block\01_Th+Ex.p65

The solution on evaporation and cooling deposits yellow crystals of hydrated ferric chloride. $FeCl_3.6H_2O$

Properties

(i) $Fe_2Cl_6 \underset{}{\overset{750°C}{\rightleftharpoons}} 2FeCl_3 \overset{\text{Above } 750°C}{\rightleftharpoons} 2FeCl_2 + Cl_2$

(ii) 

(iii) It dissolves in water. The solution is acidic in nature due to its hydrolysis as shown below :

$FeCl_3 + 3HOH \rightleftharpoons Fe(OH)_3 + 3HCl$

(iv) $FeCl_3 + 6NH_3 \rightarrow FeCl_3.6NH_3$

(v) Ferric chloride acts as an oxidising agent.

(a) It oxidies stannous chloride to stannic chloride.

$2FeCl_3 + SnCl_2 \rightarrow 2FeCl_2 + SnCl_4$

(b) It oxidises SO_2 to H_2SO_4

$2FeCl_3 + SO_2 + 2H_2O \rightarrow 2FeCl_2 + H_2SO_4 + 2HCl$

(c) It oxidises H_2S to S

$2FeCl_3 + H_2S \rightarrow 2FeCl_2 + 2HCl + S$

(d) It liberates iodine from KI.

$2FeCl_3 + 2KI \rightarrow 2FeCl_2 + 2KCl + I_2$

(e) Nascent hydrogen reduces $FeCl_3$ into $FeCl_2$

$FeCl_3 + [H] \rightarrow FeCl_2 + HCl$

(vi) $FeCl_3 + 3NH_4OH \rightarrow Fe(OH)_3 + 3NH_4Cl$

(vii) $FeCl_3 + NH_4SCN \rightarrow Fe(SCN)Cl_2 + NH_4Cl$

or $FeCl_3 + 3NH_4SCN \rightarrow Fe(SCN)_3 + 3NH_4Cl$

(viii) $4FeCl_3 + 3K_4Fe(CN)_6 \rightarrow Fe_4[Fe(CN)_6]_3 + 12KCl$

Prussian blue

(Ferri ferrocyanide)

(ix) On heating hydrated ferric chloride $FeCl_3.6H_2O$, anhydrous ferric chloride is not obtained. It is changed to Fe_2O_3 with evolution of H_2O and HCl.

$2[FeCl_3.6H_2O] \xrightarrow{\text{Heat}} Fe_2O_3 + 6HCl + 9H_2O$

Hydrated ferric chloride may be dehydrated by heating with thionyl chloride.

$FeCl_3.6H_2O + 6SOCl_2 \longrightarrow FeCl_3 + 12HCl + 6SO_2$

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\d-Salt Anal-Heating Eff+s_d-Block-Ron of Salt Anal-(OC)\Eng\04_d-Block\01_Th+Ex.p65